

2024

Time : 3 hours

Full Marks : 70

Pass Marks : 32

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Answer any five questions, in which

Q. No. 1 is compulsory.

1. Answer any four of the following : $3\frac{1}{2} \times 4 = 14$

(a) Explain gyromagnetic ratio.

(b) Define space quantization of the spin angular momentum.

(c) What do you mean by equivalent and non-equivalent electrons ?

- (d) Write down the feature of Raman-spectra.
- (e) Define the term Q-switching of Laser.
- (f) Draw the energy level of a three-level laser system.
2. Describe the S. G. experiment and discuss its importance. $10+4 = 14$
3. Obtain the energy correction due to quantum mechanical relativistic correction to the hydrogen atom. 14
4. Find the spectroscopic terms of : $4+10 = 14$
- (a) s^1p^1
- (b) p^2
5. Give the detail theory of Paschen – Back effect with a suitable example. $8+6 = 14$
6. Discuss the electronic spectra of diatomic molecules. 14

7. What are the basic principles of X-ray ?
Give the theory of X-ray emission and absorption.

4+10 = 14

8. Describe with a neat diagram, the construction and working of He-Ne laser.

14

9. Write short notes on any two of the following :

7×2 = 14

- (a) Laser Rate equation
- (b) Lande's g-factor
- (c) Frank-Condon principle
- (d) ESR

